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On the Atiyah-Hitchin-Drinfeld-Manin Vanishing Theorem for Cohomology Groups of Instanton Bundles

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1.

Let $\pi: P \to S^4$ be a principal bundle with structure group G a compact Lie group. A connection a in P is said to be self-dual if $*\Omega = \Omega$ where Ω is the curvature 2-form of a and $*\Lambda^2 \to \Lambda^2$ the Hodge duality map (for the standard metric and orientation on S^4). Such connections represent Euclidean Yang-Mills potentials with minimal action, [3] (with suitable decay at infinity, guaranteeing an extension to the compactification S^4).

If a unitary representation of G is chosen, we obtain a Hermitian vector bundle F with a connection

$$D: \Gamma(F) \to \Gamma(\Lambda^1 \otimes F)$$

(all connections are assumed to be metric), whose curvature

$$\Omega \in \Gamma(\Lambda^2 \otimes \operatorname{End} F)$$

satisfies $*\Omega = \Omega$. Conversely, by considering the linear holonomy group, a Hermitian vector bundle with a connection satisfying $*\Omega = \Omega$ gives rise to a principal bundle with self-dual connection. In this paper we consider these connections from the vector bundle point of view.

The bundle F is determined, up to isomorphism, by its rank and its second Chern class $c_2(F)$, which may be identified with an integer $k > 0$. Since S^4 has no second cohomology, it follows $c_1(F) = 0$.

S^4 may be identified with $P^1(\mathbb{H})$, the quaternion projective line. If we identify $\mathbb{H}$ with $\mathbb{C}^2$ by $\mathbb{H} = \mathbb{C} + j\mathbb{C}$, then $\mathbb{H}^2 = \mathbb{C}^4$, and the inclusion $\mathbb{C} \subset \mathbb{H}$ induces a map

$$\pi: P^3(\mathbb{C}) \to P^1(\mathbb{H}) = S^4 . \tag{1}$$

This map is onto, and the fibres are complex projective lines. Let $P^3 = P^3(\mathbb{C})$ and for each x in S^4 let $i_x: \pi^{-1}(x) \to P^3$ be the inclusion map of the fibre. The fibering (1) may be described as follows: The point of P^3 with homogeneous complex coordinates $(z_1, \ldots, z_4)$ is mapped by π to the point of S^4 with homogeneous

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quaternion coordinates (q_1, q_2) where

$$q_1 = z_1 + jz_2\,, \quad q_2 = z_3 + jz_4\,. \tag{2}$$

From this it easily follows that a 2-form ω on S^4 is self-dual if and only if $\pi^*\omega$ is of type $(1,1)$ on P^3.

This means that if $E = \pi^*F$ is the pull-back of F to P^3, and E is given the induced connection $\tilde{D}$ which has curvature $\tilde{\Omega} = \pi^*\Omega$, then D is self-dual if and only if $\tilde{\Omega}$ is of type $(1,1)$. Let $\tilde{D} = D' + D''$ be the decomposition of $\tilde{D}$ into parts of type $(1,0)$ and $(0,1)$, respectively, so that

$$D'' : \Gamma(E) \to \Gamma(\Lambda^{0,1} \otimes E) \tag{3}$$

is flat precisely when D is self-dual. Moreover D may be recovered from D'' using the metric property.

But a bundle E with an operator D'' as in (3) which is flat has a unique holomorphic structure such that a section s of E on an open set U is holomorphic if and only if $D''s = 0$. D'' is uniquely determined by this condition, so the original self-dual connection has been encoded as a holomorphic structure in the bundle E. Moreover there are some further conditions on E which encode the fact that E is a pull-back of a Hermitian vector bundle F on S^4. This was the main result of [5] (see also [4]) and made possible the application of techniques of algebraic geometry to study these self-dual connections. See, for example, [7].

The bundle E has $c_1(E) = 0$, $c_2(E) = k$ and is holomorphically trivial on each fibre of π. Let H be the holomorphic line bundle on P^3 whose Chern class is the positive generator of $H^2(P^3, \mathbb{Z})$, and for any holomorphic bundle E let $E(p) = E \otimes H^p$ where $p \in \mathbb{Z}$.

The bundles E arising from self-dual connections are called instanton bundles. If the original group G is orthogonal or symplectic, E carries a holomorphic bilinear form which makes $E \cong E'$. In [6] it is shown that if $E \cong E'$, $H^1(E(-2)) = 0$ and E restricted to some line is holomorphically trivial then there is a sequence

$$kH^{-1} \xrightarrow{a} (2k + r)\underset{\sim}{\mathbb{C}} \xrightarrow{b} kH$$

of holomorphic vector bundles (kL means the direct sum of k copies of L, and $\underset{\sim}{\mathbb{C}}$ denotes the trivial bundle with fibre $\mathbb{C}$) with $b \circ a = 0$, a is injective, b is surjective and

$$E \cong \operatorname{Ker} b / \operatorname{Im} a\,.$$

Moreover $b = a'$ with respect to a non-singular bilinear form on the middle term arising from the isomorphism $E \cong E'$. Using this it was shown in [2] how to construct all self-dual connections on S^4.

In [2] it was stated that $H^1(E(-2))$ vanished because this cohomology group could be converted by the twistor transform of Penrose [9] into solutions of the equation

$$(D^*D + R/6)s = 0\,, \quad s \in \Gamma(F)\,.$$

Since D^*D is non-negative and R (the scalar curvature of the sphere) is greater than zero this equation has only the trivial solution.

The purpose of this paper is to describe an explicit bijection between

$$H^1(E(-2)|\pi^{-1}U)$$

and the solutions of

$$(D^*D+R/6)s=0\,, \qquad s\in\Gamma(F|U) \tag{4}$$

for all open sets U in S^4. Taking $U=S^4$ gives the result. The (unpublished) proof of this relationship due to the authors of [2] uses algebrogeometric methods to establish the existence of some second order differential equation and then identifies which equation using invariance arguments.

Our proof is based on identifying $H^1(E(-2)|\pi^{-1}U)$ with the corresponding Dolbeault cohomology group. The operator

$$D'':\Gamma(E)\to\Gamma(\Lambda^{0,1}\otimes E)$$

gives rise also to an operator

$$D'':\Gamma(E(p))\to\Gamma(\Lambda^{0,1}\otimes E(p))$$

which extends to give a complex

$$\Gamma(E(p))\xrightarrow{D''}\Gamma(\Lambda^{0,1}\otimes E(p))\xrightarrow{D''}\Gamma(\Lambda^{0,2}\otimes E(p))\xrightarrow{D''}\Gamma(\Lambda^{0,3}\otimes E(p))\,.$$

The Dolbeault cohomology groups are the cohomology groups of this complex and this complex may be restricted to any open subset $\pi^{-1}U$ to give a complex which computes the cohomology groups of $E(p)|\pi^{-1}U$.

The idea of our proof is to split the equation

$$D''\alpha=0$$

for α in $\Gamma(\Lambda^{0,1}\otimes E(-2)|\pi^{-1}U)$ into horizontal and vertical parts relative to the fibering (1). It turns out that the horizontal part of α is completely determined in terms of the vertical part. If α is required to be harmonic when pulled back to every fibre, then the vertical part of α determines a section s of F on U which satisfies (4). The map $\alpha\mapsto s$ is a bijection on cohomology, so establishes the result.

Finally we remark that, using (2), the open sets U_ε in S^4 given by

$$|q_1q_2^{-1}|^2<\varepsilon$$

have inverse images in P^3 given by

$$|z_1|^2+|z_2|^2-\varepsilon|z_3|^2-\varepsilon|z_4|^2<0\,,$$

and these sets are holomorphically 1-complete [1, 10]. The open sets U_ε form a basis for the neighbourhoods of the point $q_1=0$, and by making a suitable quaternion-linear transformation we see that every point of S^4 has a system of neighbourhoods U such that $\pi^{-1}U$ is holomorphically 1-complete.

It follows that for any holomorphic bundle E on P^3 the higher direct images $R^q\pi_*\mathcal{O}(E)$ vanish for $q\geqq 2$. For instanton bundles E we also have $\pi_*\mathcal{O}(E(p))=0$ for $p<0$, and so the only direct image of $\mathcal{O}(E(p))$ which may be non-zero for $p<0$ is $R^1\pi_*\mathcal{O}(E(p))$. What we have shown above is that for $p=-2$, $R^1\pi_*\mathcal{O}(E(-2))$ is isomorphic to the sheaf-theoretic kernel of $D^*D+R/6$.

2.

Let us begin by reviewing the relationship between E and F. The pull-back of a bundle F on S^4 to P^3 is defined by

$$E=\{(p,v)\in P^3\times F|\pi p=\pi_F v\}\ ,$$

where $\pi_F:F\to S^4$ is the projection in the bundle F, and $\pi_E:E\to P^3$ is given by

$$\pi_E(p,v)=p\ .$$

If $s:U\to F$ is a local section of F, and U is an open subset of S^4, then there is a section π^*s of E over $\pi^{-1}U$ given by

$$\pi^*s(p)=(p,s(\pi p))\ .$$

Then a local frame $(s_1,\ldots,s_N)$ for F gives rise to a local frame $(\pi^*s_1,\ldots,\pi^*s_N)$ for E showing E is locally trivial and hence a vector bundle.

Suppose that

$$D:\Gamma(F)\to\Gamma(\Lambda^1\otimes F)$$

is a connection, then since E is locally framed by sections of the form π^*s, we can define $\tilde{D}$ in E by

$$\tilde{D}\pi^*s=\pi^*Ds\ .$$

Then it is clear that the curvature $\tilde{\Omega}$ of $\tilde{D}$ is the pull-back of the curvature Ω of D.

Similarly if F has a Hermitian structure, which we can regard as a map $|s|^2$ from $\Gamma(F)$ to $C^\infty(S^4)$, then E inherits a Hermitian structure by setting

$$|\pi^*s|^2=|s|^2\circ\pi\ .$$

Then a simple calculation shows that $\tilde{D}$ is a metric connection if D is.

Observe that if E is restricted to a fibre $\pi^{-1}x$ it is isomorphic to the product bundle $\pi^{-1}x\times F_x$, that the induced connection is the trivial flat connection and the induced Hermitian structure is constant.

If the curvature Ω of D is self-dual then $\tilde{\Omega}$ is of type $(1,1)$, and hence has no component of type $(0,2)$. Let D'' be the part of $\tilde{D}$ of type $(0,1)$, so that D'' is an operator

$$D'':\Gamma(E)\to\Gamma(\Lambda^{0,1}\otimes E)\ ,$$

then $(D'')^2$ is a section of

$$\Lambda^{0,2}\otimes\operatorname{End}E$$

which one readily verifies is the $(0,2)$ component of $\tilde{\Omega}$. Since this is zero we have $(D'')^2=0$ and hence obtain a complex

$$\Gamma(E)\xrightarrow{D''}\Gamma(\Lambda^{0,1}\otimes E)\xrightarrow{D''}\Gamma(\Lambda^{0,2}\otimes E)\xrightarrow{D''}\Gamma(\Lambda^{0,3}\otimes E)\ . \tag{5}$$

It follows that E has a unique holomorphic structure such that $s:U\to E$ is holomorphic if and only if $D''s=0$ where U is an open set in P^3. In this case (5)

becomes the Dolbeault complex of E and the cohomology groups of (5) are isomorphic to the cohomology groups $H^q(E)$ of the sheaf $\mathcal{O}(E)$ of germs of holomorphic sections of E. For any holomorphic bundle E on P^3 we shall take $H^q(E)$ to mean the q-th cohomology group of the complex (5).

If H is the holomorphic line bundle on P^3 defined in Sect. 1, and H^p is its p-th tensor power, $E(p)$ denotes $E\otimes H^p$ and the latter also has an operator

$$D'' : \Gamma(E(p)) \to \Gamma(\Lambda^{0,1}\otimes E(p))$$

and a corresponding complex. The cohomology groups $H^q(E(p))$ are called the twisted cohomology groups of E. We are primarily interested in $H^1(E(-2))$.

If g denotes the standard Hermitian metric on P^3, and $E(p)$ is given the Hermitian structure from those of E and H^p, then the formal adjoint $(D'')^*$ of D'' may be defined, and an element α of $\Gamma(\Lambda^{0,q}\otimes E(p))$ is called harmonic if $D''\alpha$ and $(D'')^*\alpha$ both vanish. It is known that every cohomology class in $H^q(E(p))$ has a unique harmonic representative. It turns out that a class in $H^1(E(-2))$ has another representative which is more suitable for our purposes.

The Hermitian metric on P^3 induces a metric on each fibre of π. Let

$$i_x : \pi^{-1}x \hookrightarrow P^3$$

be the inclusion map of the fibre. We shall call an element α of $\Gamma(\Lambda^{0,1}\otimes E(-2)|\pi^{-1}U)$ fibre-harmonic if $i_x^*\alpha$ is harmonic for all x in U. Here U is an open subset of S^4 which will remain fixed from now on, and for simplicity we suppress it from the notation. Let $\Gamma_{fh}(\Lambda^{0,1}\otimes E(-2))$ be the space of fibre-harmonic elements of $\Gamma(\Lambda^{0,1}\otimes E(-2))$.

We define the horizontal subbundle $H^{0,1}$ of $\Lambda^{0,1}$ to be all $(0,1)$ cotangents which vanish on tangents to the fibres of π. Let $V^{0,1}$ be the orthogonal complement of $H^{0,1}$ with respect to g. Then

$$\Lambda^{0,1} = H^{0,1}\oplus V^{0,1} ,$$

and we call $V^{0,1}$ the vertical cotangents of type $(0,1)$. Restricting $V^{0,1}$ to a fibre $\pi^{-1}x$ gives the bundle $\Lambda^{0,1}(x)$ of cotangents of type $(0,1)$ of the fibre. $\Lambda^{0,1}(x)$ is dual to the canonical bundle of $\pi^{-1}x$ which is isomorphic to $H^{-2}|\pi^{-1}x$. It follows that $V^{0,1}\otimes H^{-2}$ is trivial on the fibres of π and so is the pull-back of a line bundle on S^4. But all the line bundles on S^4 are trivial and hence $V^{0,1}\otimes H^{-2}$ is globally (C^∞) trivial on P^3.

It follows that $V^{0,1}\otimes H^{-2}$ admits a nowhere vanishing section. In Sect. 3 we define such a section σ with the additional property that σ is fibre-harmonic. If α is in $\Gamma(\Lambda^{0,1}\otimes E(-2))$ then it may be written

$$\alpha = \alpha_1 + \alpha_2$$

with

$$\alpha_1 \in \Gamma(H^{0,1}\otimes E(-2)) , \qquad \alpha_2 \in \Gamma(V^{0,1}\otimes E(-2)) .$$

But we necessarily have

$$\alpha_2 = t\otimes\sigma$$

with t a section of E. Since α_1 vanishes when pulled back to any fibre, we have

$$i_x^*\alpha = i_x^*\alpha_2 = t|\pi^{-1}x \otimes \sigma|\pi^{-1}x \ .$$

Then $i_x^*\alpha$ is harmonic if and only if $t|\pi^{-1}x$ is constant as a section of $E|\pi^{-1}x = \pi^{-1}x \times F_x$. That is: if α_2 is the vertical part of a fibre-harmonic form α then $\alpha_2 = \pi^* s \otimes \sigma$ for a section s of F on U. We have therefore shown:

Proposition 1. *The fibre-harmonic elements α of $\Gamma(\Lambda^{0,1} \otimes E(-2))$ have the form*

$$\alpha = \alpha_1 + \pi^* s \otimes \sigma \tag{6}$$

with s in $\Gamma(F)$ and α_1 in $\Gamma(H^{0,1} \otimes E(-2))$ arbitrary.

Proposition 2. $\Gamma(\Lambda^{0,1} \otimes E(-2)) = \Gamma_{fh}(\Lambda^{0,1} \otimes E(-2)) \oplus D''\Gamma(E(-2))$.

Proof. See Sect. 3.

Thus any α in $\Gamma(\Lambda^{0,1} \otimes E(-2))$ can be written in the form

$$\alpha = \alpha_1 + \pi^* s \otimes \sigma + D''t \tag{7}$$

with α_1, s and t unique. We define, for $[\alpha]$ in $H^1(E(-2))$,

$$\pi_*[\alpha] = s$$

if α decomposes as in (7).

Proposition 3. $\pi_* : H^1(E(-2)) \to \Gamma(F)$ *is an injection.*

Proof. This follows from the next proposition.

Proposition 4. *If α in $\Gamma(H^{0,1} \otimes E(-2))$ satisfies $D''\alpha = 0$ then $\alpha = 0$.*

Proof. See Sect. 3.

In order to identify the image of π_* we now have to see what constraint

$$D''[\alpha_1 + \pi^* s \otimes \sigma] = 0 \tag{8}$$

imposes on s if α_1 is in $\Gamma(H^{0,1} \otimes E(-2))$. Let us observe that the decomposition

$$\Lambda^{0,1} = H^{0,1} \oplus V^{0,1}$$

implies

$$\Lambda^{0,2} = \Lambda^2 H^{0,1} \oplus H^{0,1} \otimes V^{0,1}$$

and that

$$\Lambda^{0,3} = \Lambda^2 \mathrm{H}^{0,1} \otimes \mathrm{V}^{0,1} \ .$$

Now $\Lambda^{0,3}$ is dual to the canonical bundle of P^3 and so $\Lambda^{0,3} \cong H^4$ as C^∞ vector bundles. Since $V^{0,1} \cong H^2$ it follows we also have $\Lambda^2 H^{0,1} \cong H^2$ and hence $\Lambda^2 H^{0,1} \otimes H^{-2}$ admits a nowhere vanishing section. In Sect. 3 we construct such a section τ and find that

$$\bar{\partial}\sigma = 2\tau \ .$$

Instead of (8) we shall study the weaker condition

$$D''(\alpha_1+\pi^*s\otimes\sigma)\in\Gamma(\Lambda^2H^{0,1}\otimes E(-2)) . \tag{9}$$

This says that the component of $D''(\alpha_1+\pi^*s\otimes\sigma)$ in $\Gamma(H^{0,1}\otimes V^{0,1}\otimes E(-2))$ must vanish. Since $D''\pi^*s$ is in $\Gamma(H^{0,1}\otimes E)$, and $\bar\partial\sigma$ is in $\Gamma(\Lambda^2H^{0,1}\otimes H^{-2})$, this is equivalent to the requirement that the component of $D''\alpha_1$ in $\Gamma(H^{0,1}\otimes V^{0,1}\otimes E(-2))$ is $-D''(\pi^*s\otimes\sigma)$. Let

$$D_1:\Gamma(H^{0,1}\otimes E(-2))\to\Gamma(H^{0,1}\otimes V^{0,1}\otimes E(-2))$$

be such that $D_1\alpha$ is the component of $D''\alpha$ in $\Gamma(H^{0,1}\otimes V^{0,1}\otimes E(-2))$. Restricted to a fibre $\pi^{-1}x$, D_1 determines a holomorphic structure in $H^{0,1}\otimes E(-2)|\pi^{-1}x$, whilst as C^∞ bundles

$$H^{0,1}\otimes E(-2)|\pi^{-1}x\cong F_x\otimes 2H\otimes H^{-2}\cong 2F_x\otimes H^{-1} .$$

In fact D_1 determines the natural holomorphic structure in $2F_x\otimes H^{-1}$. But H^{-1} has all its cohomology groups equal to zero and so D_1 is a linear isomorphism. That is:

Proposition 5. *Given s in $\Gamma(F)$ there is a unique $\phi(s)$ in $\Gamma(H^{0,1}\otimes E(-2))$ with*

$$D_1\phi(s)=-D''\pi^*s\otimes\sigma .$$

Proof. See Sect. 3.

For s in $\Gamma(F)$ we define

$$\psi(s)=\phi(s)+\pi^*s\otimes\sigma$$

so that

$$D''\psi(s)=t\otimes\tau$$

for some section t of E. Applying D'' to both sides, and using $\bar\partial\tau=0$, we find

$$D''t\wedge\tau=0 ,$$

and hence

$$D''t\in\Gamma(H^{0,1}\otimes E) .$$

This says that t is holomorphic when restricted to each fibre, but since E is constant on each fibre and the fibres are compact, t is constant on the fibres and therefore of the form π^*t' for some section t' of F. In fact t' can be computed explicitly to give:

Proposition 6. $D''\psi(s)=\pi^*[(D^*D+2)s]\otimes\tau$.

Proof. See Sect. 3.

Finally, for the metric we are using on S^4, $R=12$ and so we have:

Theorem. $\pi_*:H^1(E(-2)|\pi^{-1}U)\to\Gamma(F|U)$ *maps* $H^1(E(-2)|\pi^{-1}U)$ *bijectively onto the kernel of* $D^*D+R/6$.

Proof. By Proposition 2 any class $[\alpha]$ in $H^1(E(-2)|\pi^{-1}U)$ has a unique fibre-harmonic representative α which by Proposition 1 has the form $\alpha=\alpha_1+\pi^*s\otimes\sigma$ with α_1 in $\Gamma(H^{0,1}\otimes E(-2))$. Since $D''\alpha=0$ condition (9) is certainly satisfied, so $\alpha_1=\phi(s)$ and hence $\alpha=\psi(s)$. Then from Proposition 6 it follows $(D^*D+2)s=0$. Conversely, if $(D^*D+2)s=0$ then Proposition 6 implies $D''\psi(s)=0$. But $\psi(s)=\phi(s)+\pi^*s\otimes\sigma$ means $\pi_*[\psi(s)]=s$ and hence π_* is onto. It is injective by Proposition 3.

3.

In this section we shall give proofs of the propositions of the previous section. We begin with Proposition 2.

Let α be in $\Gamma(\Lambda^{0,1}\otimes E(-2))$ then $i_x^*\alpha$ is in $\Gamma(\Lambda^{0,1}(x)\otimes E(-2)|\pi^{-1}x)$, and so is closed. Then by the Hodge-Kodaira theory it decomposes into a sum of a harmonic and an exact term. Thus

$$i_x^*\alpha=\beta_x+\bar\partial_x t_x\,,$$

where t_x is a section of $E(-2)|\pi^{-1}x$. β_x is unique, and since $E(-2)|\pi^{-1}x$ has no holomorphic sections t_x is also unique. Thus t_x depends smoothly on x, see [8], and we obtain a section t of $E(-2)$ such that

$$t|\pi^{-1}x=t_x\,.$$

Then consider $\alpha-D''t$. Since $i_x^*D''t=\bar\partial_x t_x$ it follows $\alpha-D''t$ is fibre-harmonic, so

$$\Gamma(\Lambda^{0,1}\otimes E(-2))=\Gamma_{fh}(\Lambda^{0,1}\otimes E(-2)+D''\Gamma(E(-2))\,.$$

To show the sum is direct, suppose

$$0=\beta+D''t$$

with β in $\Gamma_{fh}(\Lambda^{0,1}\otimes E(-2))$ and t in $\Gamma(E(-2))$. Applying i_x^* we obtain

$$0=i_x^*\beta+\bar\partial_x t_x\,,$$

and hence

$$i_x^*\beta=0=\bar\partial_x t_x\,,$$

because a harmonic form which is exact must vanish. But $E(-2)|\pi^{-1}x$ has no holomorphic sections and so t_x vanishes for each x. Hence $t=0=\beta$.

Proposition 4 is a special case of Proposition 5 by taking $s=0$ in (9). Hence we must study the restriction to the fibre of the part of D'' going from $\Gamma(H^{0,1}\otimes E(-2))$ to $\Gamma(H^{0,1}\otimes V^{0,1}\otimes E(-2))$. This is accomplished by choosing suitable coordinates on P^3 and S^4.

Let V_n, $n=1,2$, be the open subsets of S^4 where $q_n\neq 0$ and U_i, $i=1,2,3,4$, the open subsets of P^3 where $z_i\neq 0$. Then

$$\pi^{-1}V_1=U_1\cup U_2\,,\qquad \pi^{-1}V_2=U_3\cup U_4\,.$$

On V_n define x_n in $\mathbb{H}$ by

$$x_1 = q_2 q_1^{-1}, \quad x_2 = q_1 q_2^{-1}$$

so that $x_1 = x_2^{-1}$ on $V_1 \cap V_2$. We regard x_n as a vector in $\mathbb{R}^4$ by means of the standard basis $1, i, j, k$, for $\mathbb{H}$:

$$x_n = x_n^1 1 + x_n^2 i + x_n^3 j + x_n^4 k ,$$

and we set

$$x_n = u_n + j v_n$$

with

$$u_n = x_n^1 + i x_n^2 , \quad v_n = x_n^3 - i x_n^4$$

being complex numbers. Let

$$\partial_\mu^n = \partial/\partial x_n^\mu , \quad \partial_u^n = \tfrac{1}{2}(\partial_1^n - i\partial_2^n) , \quad \partial_v^n = \tfrac{1}{2}(\partial_3^n + i\partial_4^n) ,$$
$$\partial_{\bar{u}}^n = \overline{\partial_u^n} , \quad \partial_{\bar{v}}^n = \overline{\partial_v^n} .$$

The open sets V_n are contractible, so there are orthonormal frame fields $b_n = (s_n^1, \ldots, s_n^N)$ for F on V_n and

$$b_2 = b_1 \cdot c_{12}$$

on $V_1 \cap V_2$ with

$$c_{12} : V_1 \cap V_2 \to U(N) .$$

Any section s of F on an open set U can be written

$$s | U \cap V_n = \sum_a f_n^a s_n^a = b_n \cdot f_n$$

with

$$f_n : U \cap V_n \to \mathbb{C}^N$$

and

$$f_1 = c_{12} f_2$$

on $V_1 \cap V_2 \cap U$.

If D is the connection in F let

$$D s_n^b = \sum_a \theta_n^{ab} \otimes s_n^a$$

define the matrix θ_n of connection forms so that

$$Ds | U \cap V_n = b_n \cdot (df_n + \theta_n f_n) .$$

For any vector field X on V_n let

$$D_X^n = X + \theta_n(X) .$$

On P^3 we have the usual affine coordinates $z_j/z_i, j \neq i$ on U_i: We label these as follows:

On U_1, $\xi_1^1 = z_3/z_1$, $\xi_2^1 = z_4/z_1$, $\xi_3^1 = z_2/z_1$;
on U_2, $\xi_1^2 = z_3/z_2$, $\xi_2^2 = z_4/z_2$, $\xi_3^2 = z_1/z_2$;
on U_3, $\xi_1^3 = z_1/z_3$, $\xi_2^3 = z_2/z_3$, $\xi_3^3 = z_4/z_3$;
on U_4, $\xi_1^4 = z_1/z_4$, $\xi_2^4 = z_2/z_4$, $\xi_3^4 = z_3/z_4$.

Put

$$n_1 = n_2 = 1 \qquad n_3 = n_4 = 2,$$

then

$$\xi_1^i + j\xi_2^i = x_{n_i} q_{n_i} z_i^{-1}.$$

Let

$$\eta_i = \xi_3^i.$$

Then on U_i we can take either ξ_1^i, ξ_2^i, ξ_3^i as coordinates or u_{n_i}, v_{n_i}, η_i as (non-holomorphic) coordinates. The former are useful for computing $\bar{\partial}$ whilst the latter are adapted to the fibering. We have the relations

$$\xi_1^i + j\xi_2^i = \begin{cases} x_{n_i}(1 + j\eta_i), & i = 1, 3, \\ x_{n_i}(\eta_i + j), & i = 2, 4. \end{cases}$$

Thus

$$\xi_1^i = u_{n_i} - \bar{v}_{n_i}\eta_i, \quad \xi_2^i = v_{n_i} + \bar{u}_{n_i}\eta_i, \quad i = 1, 3,$$
$$\xi_1^i = u_{n_i}\eta_i - \bar{v}_{n_i}, \quad \xi_2^i = v_{n_i}\eta_i + \bar{u}_{n_i}, \quad i = 2, 4.$$

Let us define

$$\alpha_1^i = \lambda_i(d\bar{u}_{n_i} - \bar{\eta}_i dv_{n_i}), \quad \alpha_2^i = \lambda_i(d\bar{v}_{n_i} + \bar{\eta}_i du_{n_i}), \quad i = 1, 3,$$
$$\alpha_1^i = \lambda_i(\bar{\eta}_i d\bar{u}_{n_i} - dv_{n_i}), \quad \alpha_2^i = \lambda_i(\bar{\eta}_i d\bar{v}_{n_i} + du_{n_i}), \quad i = 2, 4.$$

where

$$\lambda_i = (1 + |x_{n_i}|^2)^{-2}(1 + |\eta_i|^2)^{-1}.$$

Then by construction, (α_1^i, α_2^i) forms a local frame field for $H^{0,1}$ on U_i. In the coordinates (u, v, η) we define

$$\alpha_3^i = (1 + |x_{n_i}|^2)^{-1}\partial/\partial\eta_i \lrcorner g$$

on U_i. In terms of ξ_1^i, ξ_2^i, ξ_3^i we have

$$g = \frac{|d\bar{\xi}_1^i|^2 + |d\bar{\xi}_2^i|^2 + |d\bar{\xi}_3^i|^2}{1 + |\xi_1^i|^2 + |\xi_2^i|^2 + |\xi_3^i|^2} - \frac{|\xi_1^i d\bar{\xi}_1^i + \xi_2^i d\bar{\xi}_2^i + \xi_3^i d\bar{\xi}_3^i|^2}{(1 + |\xi_1^i|^2 + |\xi_2^i|^2 + |\xi_3^i|^2)^2}$$

on U_i, from which explicit expressions for the α_3^i may be found. For example,

$$\alpha_3^1 = \frac{\bar\xi_1^1 d\bar\xi_2^1 - \bar\xi_2^1 d\bar\xi_1^1 + d\bar\xi_3^1}{(1+|\xi_1^1|^2+|\xi_2^1|^2+|\xi_3^1|^2)^2} = \frac{\bar\xi_1^1\alpha_2^1 - \bar\xi_2^1\alpha_1^1}{1+|\eta_1|^2} + \frac{d\bar\eta_1}{(1+|x_1|^2)(1+|\eta_1|^2)^2} .$$

The reader may check that

$$\bar\partial\alpha_1^i = 2\xi_2^i(1+|x_{n_i}|^2)\alpha_1^i \wedge \alpha_2^i ,$$
$$\bar\partial\alpha_2^i = -2\xi_1^i(1+|x_{n_i}|^2)\alpha_1^i \wedge \alpha_2^i ,$$
$$\bar\partial\alpha_3^i = \pm 2(1+|x_{n_i}|^2)\alpha_1^i \wedge \alpha_2^i ,$$

with the plus sign on U_1, U_3, the minus sign on U_2, U_4.

α_3^i is a basis for $V^{0,1}$ on U_i. Let Y_1^i, Y_2^i, Y_3^i be the dual basis to α_1^i, α_2^i, α_3^i for the tangents of type $(0,1)$ on U_i. If

$$X_1^i = (1+|x_{n_i}|^2)^2(\partial_{\bar u}^{n_i} - \eta_i\partial_v^{n_i}) ,$$
$$X_2^i = (1+|x_{n_i}|^2)^2(\partial_{\bar v}^{n_i} + \eta_i\partial_u^{n_i}) , \qquad i=1,3 ,$$
$$X_1^i = (1+|x_{n_i}|^2)^2(\eta_i\partial_{\bar u}^{n_i} - \partial_v^{n_i}) ,$$
$$X_2^i = (1+|x_{n_i}|^2)^2(\eta_i\partial_{\bar v}^{n_i} + \partial_u^{n_i}) , \qquad i=2,4 ,$$

then

$$Y_1^i = X_1^i + (1+|x_{n_i}|^2)(1+|\eta_i|^2)\bar\xi_2^i\partial/\partial\bar\eta_i ,$$
$$Y_2^i = X_2^i - (1+|x_{n_i}|^2)(1+|\eta_i|^2)\bar\xi_1^i\partial/\partial\bar\eta_i ,$$
$$Y_3^i = (1+|x_{n_i}|^2)(1+|\eta_i|^2)^2\partial/\partial\bar\eta_i .$$

The bundle H may be described relative to the covering U_i as follows: H has a nowhere vanishing holomorphic section s_i on U_i such that on $U_i \cap U_j$

$$s_j = (z_j/z_i)s_i .$$

Then H^p has a section s_i^p on U_i and on $U_i \cap U_j$

$$s_j^p = (z_j/z_i)^p s_i .$$

It follows that on U_i any element α of $\Gamma(\Lambda^{0,1}\otimes E(-2))$ can be written

$$\alpha = (\alpha_1^i \otimes \tilde b_i \cdot f_i^1 + \alpha_2^i \otimes \tilde b_i \cdot f_i^2 + \alpha_3^i \otimes \tilde b_i \cdot f_i^3) \otimes s_i^{-2} , \tag{10}$$

where for brevity we let $\tilde b_i = \pi^* b_{n_i}$ be the pull-back to E of the local frame b_{n_i} for F. Then, since s_i is holomorphic,

$$\begin{aligned} D''\alpha = \{ & \alpha_1^i \wedge \alpha_3^i \otimes \tilde b_i \cdot [\tilde D_1^i(f_i^3) - Y_3^i(f_i^1)] \\ & + \alpha_2^i \wedge \alpha_3^i \otimes \tilde b_i \cdot [\tilde D_2^i(f_i^3) - Y_3^i(f_i^2)] \\ & + \alpha_1^i \wedge \alpha_2^i \otimes \tilde b_i \cdot [\tilde D_1^i(f_i^2) - \tilde D_2^i(f_i^1) + 2(1+|x_{n_i}|^2)(\xi_2^i f_i^1 - \xi_1^i f_i^2 + 2f_i^3)]\} \otimes s_i^{-2} \end{aligned}$$

on U_i, where

$$\tilde D_n^i = Y_n^i + \pi^*\theta_{n_i}(Y_n^i) , \qquad i=1,\ldots,4 , \quad n=1,2 .$$

Now we check what happens on $U_i \cap U_j$. There are too many cases to list here, but we shall check two cases which are typical. On $U_1 \cap U_2$ the neighbourhood V_1 in S^4 remains fixed, and we have

$$\alpha_1^1 = \eta_1^{-1}\alpha_1^2, \quad \alpha_2^1 = \eta_1^{-1}\alpha_2^2, \quad \alpha_3^1 = -\eta_1^{-2}\alpha_3^2,$$
$$\tilde{b}_1 = \tilde{b}_2, \quad s_1 = \eta_1 s_2 .$$

Hence

$$f_1^1 = \eta_1^{-1} f_2^1, \quad f_1^2 = \eta_1^{-1} f_2^2, \quad f_1^3 = -f_2^3$$

on $U_1 \cap U_2$. Thus we see explicitly that along a fixed fibre, sections of $H^{0,1} \otimes H^{-2}$ transform as sections of $2H^{-1}$.

If in (10) we set $f_3^i = 0$, we get a general section α of $H^{0,1} \otimes E(-2)$. Then the component of $D''\alpha$ in $\Gamma(H^{0,1} \otimes V^{0,1} \otimes E(-2))$ is

$$D_1\alpha = -(1+|x_{n_i}|^2)(1+|\eta_i|^2)^2 [\alpha_1^i \wedge \alpha_3^i \otimes \tilde{b}_i \cdot \partial f_i^1/\partial\bar{\eta}_i + \alpha_2^i \wedge \alpha_3^i \otimes \tilde{b}_i \cdot \partial f_i^2/\partial\bar{\eta}_i] \otimes s_i^{-2} \tag{12}$$

on U_i. Making a substitution $\eta_1 = \eta_2^{-1}$ we see that $D_1\alpha$ transforms as a section of $2F_x \otimes \Lambda^{0,1}(x) \otimes H^{-1}$ on $\pi^{-1}x$ for each fixed x, and that f_i^1, f_i^2 transform as sections of $2H^{-1}$ with D_1 precisely the $\bar{\partial}$ operator in this bundle. Since $\bar{\partial}$ in H^{-1} on $P^1(\mathbb{C})$ has no kernel or cokernel, it follows

$$D_1 : \Gamma(H^{0,1} \otimes E(-2)) \to \Gamma(H^{0,1} \otimes V^{0,1} \otimes E(-2))$$

is a linear isomorphism as claimed, and Proposition 5 is proven.

To define τ and σ we must see how the forms α_1^i, α_2^i, α_3^i transform with respect to coordinate changes which change the base variables x_{n_i}. For example on $U_1 \cap U_4$ we have

$$\alpha_1^4 = -\xi_2^1(u_1\alpha_1^1 + v_1\alpha_2^1), \quad \alpha_2^4 = \xi_2^1(\bar{v}_1\alpha_1^1 - \bar{u}_1\alpha_2^1),$$
$$s_4 = \xi_2^1 s_1, \quad \alpha_3^4 = -(\xi_2^1)^2\alpha_3^1, \quad 1+|\eta_4|^2 = |x_1|^2(1+|\eta_1|^2)|\xi_2^1|^{-2} .$$

Thus

$$(1+|x_2|^2)\alpha_1^4 \wedge \alpha_2^4 \otimes s_4^{-2} = (1+|x_1|^2)\alpha_1^1 \wedge \alpha_2^1 \otimes s_1^{-2}$$

and similar relations hold on the other intersections $U_i \cap U_j$. Then we may define τ by

$$\tau|U_i = (1+|x_{n_i}|^2)\alpha_1^i \wedge \alpha_2^i \otimes s_i^{-2} .$$

Similarly one finds

$$-\alpha_3^4 \otimes s_4^{-2} = \alpha_3^1 \otimes s_1^{-2}$$

on $U_1 \cap U_4$, so we set

$$\sigma|U_i = \pm\alpha_3^i \otimes s_i^{-2}$$

with the plus sign on U_1, U_3, the minus sign on U_2, U_4. An explicit calculation now shows

$$\bar{\partial}\sigma = 2\tau .$$

Note that for x in V_1

$$i_x^*\sigma=(1+|x_1|^2)^{-1}(1+|\eta_i|^2)^{-2}d\bar{\eta}_i\otimes s_i^{-2}$$

on $U_i\cap\pi^{-1}x$, $i=1,2$, and this is a basis for the harmonic elements of $\Gamma(\Lambda^{0,1}(x)\otimes H^{-2}|\pi^{-1}x)$. A similar statement holds on V_2.

In order to find an explicit formula for $\phi(s)$ in Proposition 5, we must give the component of $-D''(\pi^*s\otimes\sigma)$ lying in $\Gamma(H^{0,1}\otimes V^{0,1}\otimes E(-2))$. On U_1 it is

$$[\alpha_1^1\wedge\alpha_3^1\otimes\tilde{b}_1\cdot\tilde{D}_1^1(f_1^3)+\alpha_2^1\wedge\alpha_3^1\otimes\tilde{b}_1\cdot\tilde{D}_2^1(f_1^3)]\otimes s_1^{-2}\,,$$

where

$$f_1^3=a_1\circ\pi|U_1 \tag{13}$$

if

$$s|V_1=b_1\cdot a_1\,.$$

But then

$$\tilde{D}_1^1(f_1^3)=(1+|x_1|^2)^2(D_u^1a_1-\eta_1D_v^1a_1)\,,$$
$$\tilde{D}_2^1(f_2^3)=(1+|x_1|^2)^2(D_v^1a_1+\eta_1D_u^1a_1)\,,$$

where $D_u^1=D_{\partial_u^1}$ and so on. From (12) we have to solve

$$\partial f_1^1/\partial\bar{\eta}_1=(1+|x_1|^2)(1+|\eta_1|^2)^{-2}(D_u^1a_1-\eta_1D_v^1a_1)\,,$$
$$\partial f_1^2/\partial\bar{\eta}_1=(1+|x_1|^2)(1+|\eta_1|^2)^{-2}(D_v^1a_1+\eta_1D_u^1a_1)\,,$$

regarding f_1^1, f_1^2 as sections of H^{-1} on $P^1(\mathbb{C})$ in the η_1 variable. The unique solution is

$$\left.\begin{aligned}f_1^1&=(1+|x_1|^2)(1+|\eta_1|^2)^{-1}(D_v^1a_1+\bar{\eta}_1D_u^1a_1)\,,\\ f_1^2&=(1+|x_1|^2)(1+|\eta_1|^2)^{-1}(D_u^1a_1-\bar{\eta}_1D_v^1a_1)\,,\end{aligned}\right\} \tag{14}$$

and so we have $\phi(s)$. Substituting (13) and (14) into (11) we find

$$\begin{aligned}D''\psi(s)=\alpha_1^1\wedge\alpha_2^1\otimes\tilde{b}_1\cdot[&-D_{X_1^1}^1D_{X_1^1}^1-D_{X_2^1}^1D_{X_2^1}^1\\&+2(1+|x_1|^2)(\xi_2^1D_{X_2^1}^1+\xi_1^1D_{X_1^1}^1+2)]a_1\circ\pi\otimes s_1^{-2}\,.\end{aligned}$$

On expanding this we obtain

$$\begin{aligned}D''\psi(s)=\tilde{b}_1\cdot\Big[&-\tfrac{1}{4}(1+|x_1|^2)^2\sum_\mu D_{\partial_\mu^1}^1D_{\partial_\mu^1}^1\\&+(1+|x_1|^2)\sum_\mu x_1^\mu D_{\partial_\mu^1}^1\partial_\mu^1+2\Big]a_1\circ\pi\otimes\tau\,.\end{aligned}$$

The metric we are using on S^4 is given on V_1 by

$$4(1+|x_1|^2)^{-2}\sum_\mu(dx_1^\mu)^2\,,$$

and so

$$(D_{\partial_\mu^1}^1)^*=-\tfrac{1}{4}(1+|x_1|^2)^2D_{\partial_\mu^1}^1+(1+|x_1|^2)x_1^\mu\,,$$

so that

$$D''\psi(s)=\pi^*[(D^*D+2)s]\otimes\tau\ ,$$

and hence we obtain Proposition 6.

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